

QBITEL BRIDGE

Automotive & Connected Vehicles

Deployment Checklist

10 Phases | 135-Day Deployment | UNECE WP.29 Ready

10

Phases

135

Days

70+

Checklist Items

WP.29

Compliant

QBITEL Bridge Automotive Deployment Checklist

This checklist covers all 10 phases of a QBITEL Bridge automotive deployment, from initial WP.29 gap assessment through operational go-live. Each phase includes sign-off requirements for the responsible owner. Total deployment timeline: 135 days.

Highlight	Meaning
Standard item	Normal deployment task
Critical item	Blocks progression to next phase if incomplete
Pass/Fail item	Has binary pass/fail acceptance criteria
Warning item	Requires monitoring but does not block progression

PHASE**0****Pre-Engagement & WP.29 Gap Assessment**

Days 1-5

Owner: Automotive CISO

- ☐ Identify OEM cybersecurity lead and SCMS provider contact
- ☐ Complete UNECE WP.29 R155/R156 gap analysis
- ☐ Inventory current V2X deployment (RSU count, OBU count, SCMS provider)
- ☐ Review existing ISO/SAE 21434 TARA documentation
- ☐ Identify fleet size and OTA platform capabilities
- ☐ Define PQC algorithm selection (Falcon-512 vs Falcon-1024 vs hybrid)

Sign-off: Automotive CISO _____ Date _____**PHASE****1****V2X Protocol Discovery & Analysis**

Days 6-12

Owner: V2X Architect

- ☐ Deploy V2X passive capture on target RSU corridors
- ☐ Enumerate all V2X message types in production (BSM, SPaT, MAP, TIM, EVA)
- ☐ Audit current certificate chain (IEEE 1609.2 compliance)
- ☐ Measure baseline V2X message rates (avg/peak msg/sec)
- ☐ Assess DSRC vs C-V2X split in deployment
- ☐ Document legacy RSU firmware versions

Sign-off: V2X Architect _____ Date _____

PHASE
2**SCMS & PKI Infrastructure Readiness**

Days 13-20

Owner: PKI Team

- ☐ Confirm SCMS provider API access (CAMP or OBS)
- ☐ Provision Falcon-512 root CA in HSM (FIPS 140-3 Level 3)
- ☐ Generate pseudonym certificate pool (initial batch: 52 weeks x fleet size)
- ☐ Configure CRL distribution infrastructure (<100ms target)
- ☐ Test bulk enrollment pipeline (10,000 vehicles/hour throughput)
- ☐ Establish misbehavior reporting channel to SCMS

Sign-off: PKI Lead _____ Date _____

PHASE
3**Cryptographic Architecture Validation**

Days 21-28

Owner: Crypto Team

- ☐ Benchmark Falcon-512 verification on target ECU hardware
- ☐ Validate <5ms V2X verification latency (IEEE 1609.2 requirement: <10ms)
- ☐ Test batch verification throughput at peak density (>=1,500 msg/sec)
- ☐ Validate 666-byte implicit certificate fits DSRC frame size
- ☐ Test hybrid ECDSA+Falcon-512 dual-signing mode
- ☐ Benchmark OTA update signing (ML-DSA-65) performance

Sign-off: Crypto Lead _____ Date _____

PHASE
4**RSU Infrastructure Deployment**

Days 29-40

Owner: Infrastructure

- ☐ Update RSU firmware to QBITEL V2X authentication module
- ☐ Configure Falcon-512 certificate verification on each RSU
- ☐ Deploy CRL update mechanism on all RSUs
- ☐ Test RSU-to-RSU certificate exchange
- ☐ Validate RSU backward compatibility with legacy OBUs (ECDSA fallback)

- ☐ Monitor RSU V2X authentication success rate (target: >99.99%)

Sign-off: Infrastructure Lead _____ Date _____

PHASE

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OBU (Vehicle) Certificate Provisioning

Days 41-55

Owner: OTA Team

- ☐ Generate Falcon-512 key pairs for canary fleet (10 vehicles)
- ☐ Provision pseudonym certificate pools via SCMS API
- ☐ Deploy QBITEL V2X library via OTA to canary fleet
- ☐ Validate canary fleet V2X authentication success rate
- ☐ Monitor for regressions vs. baseline (V2X message rate, latency)
- ☐ Approve canary -> 1% fleet expansion

Sign-off: OTA Lead _____ Date _____

PHASE

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Staged Fleet OTA Rollout

Days 56-90

Owner: Fleet Operations

- ☐ Deploy to 1% of fleet (monitor 24h, error rate <0.1%)
 - ☐ Confirm automatic rollback is armed
 - ☐ Verify delta update delivery <4 hours
- ☐ Deploy to 10% of fleet (monitor 48h, error rate <0.1%)
 - ☐ Geographic diversity check
 - ☐ Latency regression check at 10% scale
- ☐ Deploy to 50% of fleet (monitor 72h)
 - ☐ Peak traffic load test during deployment
 - ☐ SCMS pseudonym pool adequacy check
- ☐ Deploy to 100% of fleet
 - ☐ Final coverage verification
 - ☐ All vehicles reporting authentication success
- ☐ Verify TPM attestation on all updated vehicles
- ☐ Generate fleet coverage report for WP.29 documentation

Sign-off: Fleet Operations Lead _____ Date _____

PHASE

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Misbehavior Detection Activation

Days 91-100

Owner: Security Team

- ☐ Enable V2X anomaly detection on all RSUs
- ☐ Configure physics-based validation (speed/position plausibility)
- ☐ Set misbehavior reporting threshold and SCMS integration
- ☐ Test spoofing detection with simulation (pass: detection <100ms)
 - ☐ Simulate 100 spoofed BSMS
 - ☐ Verify all 100 rejected within 100ms
 - ☐ Verify zero false positives on legitimate messages
- ☐ Validate false positive rate (<0.0001% of legitimate messages)
- ☐ Configure P1 alert: Active spoofing detected -> immediate SOC notification

Sign-off: Security Lead _____ Date _____

PHASE

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UNECE WP.29 / ISO 21434 Evidence Generation

Days 101-115

Owner: Compliance

- ☐ Generate automated TARA covering V2X attack surfaces
- ☐ Produce security validation test results
- ☐ Generate CSMS documentation for WP.29 R155 submission
- ☐ Produce R156 SUMS documentation and OTA audit logs
- ☐ Prepare fleet coverage certificate for type approval
- ☐ Review all evidence with legal/regulatory team

Sign-off: Compliance Lead _____ Date _____

PHASE

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Performance & Safety Validation

Days 116-125

Owner: Validation Team

- ☐ V2X latency test: 100% messages <5ms (pass/fail)
 - ☐ Test at low density (rural: 50 msg/sec)
 - ☐ Test at high density (urban: 1,000 msg/sec)
 - ☐ All results must be <5ms
- ☐ Batch throughput test: >=1,500 msg/sec at peak density (pass/fail)

☐ Spoofing detection test: 0/100 simulated spoofed messages accepted (pass/fail)

- ☐ Inject 100 crafted spoofed BSMS
- ☐ All must be rejected
- ☐ Zero legitimate messages rejected

☐ OTA rollback test: automatic rollback triggered correctly (pass/fail)

- ☐ Inject artificial error rate >0.1%
- ☐ Confirm rollback triggers automatically
- ☐ Confirm rollback completes within 30 minutes

☐ Hybrid mode test: ECDSA legacy compatibility verified (pass/fail)

- ☐ Legacy RSU sees valid ECDSA signature
- ☐ Upgraded RSU verifies both ECDSA and Falcon-512

Sign-off: Validation Lead _____ Date _____

PHASE
10

Operational Handover & Go-Live

Days 126-135

Owner: CTO / CISO

☐ Configure fleet health monitoring dashboard

☐ Establish pseudonym rotation automation (weekly)

☐ Set up 24/7 NOC alerting for V2X anomalies

☐ Complete OEM security team training (V2X security operations)

- ☐ V2X authentication monitoring
- ☐ Misbehavior detection response
- ☐ OTA rollback procedures

☐ Publish WP.29 CSMS evidence package

☐ Execute go-live sign-off

Sign-off: CTO _____ Date _____

Sign-off: CISO _____ Date _____

Appendix A: Performance Targets

Metric	Minimum Requirement	QBITEL Target	Status
V2X verification latency	<10ms	<5ms	EXCEEDS
Batch throughput	500 msg/sec	1,500 msg/sec	EXCEEDS
Certificate size	<2KB	666 bytes	EXCEEDS
CRL distribution to RSUs	<1 second	<100ms	EXCEEDS
Fleet OTA coverage	100%	Staged 12 months	COMPLIANT
False positive rate	<0.001%	<0.0001%	EXCEEDS
SCMS bulk enrollment	N/A	10,000 vehicles/hour	EXCEEDS
Cost per vehicle/year	\$50-200 (classical)	\$10-30 (PQC)	LOWER COST

Appendix B: Rollback Procedures

Automatic Rollback Trigger Conditions

- ☐ Error rate >0.1% in any deployment stage
- ☐ V2X authentication failure rate >0.01% sustained for >5 minutes
- ☐ TPM attestation failure on >10 vehicles in a batch
- ☐ OTA delivery failure rate >5% in any cohort
- ☐ V2X message latency >8ms (80% of the 10ms budget) on >1% of messages

Manual Rollback Steps

- ☐ Access fleet management console at <https://bridge.qbitel.com/fleet>
- ☐ Navigate to Fleet > OTA Management > Rollback
- ☐ Select affected vehicle cohort (by region, VIN range, or deployment batch)
- ☐ Confirm rollback to previous firmware version
- ☐ Monitor rollback completion rate (target: 100% within 30 minutes)
- ☐ Verify TPM attestation on all rolled-back vehicles
- ☐ File incident report to QBITEL NOC: noc@qbitel.com

Escalation Contacts

Severity	Contact	Response Time
P1 - Active V2X spoofing	QBITEL NOC: noc@qbitel.com +1-800-QBITEL1	<15 minutes
P2 - OTA rollback failure	QBITEL Automotive Engineering: auto@qbitel.com	<1 hour
P3 - Performance degradation	enterprise@qbitel.com	<4 hours
General enquiries	enterprise@qbitel.com bridge.qbitel.com	Next business day

enterprise@qbitel.com	bridge.qbitel.com	noc@qbitel.com (24/7)
Sales & Licensing	Portal & Documentation	NOC Support